

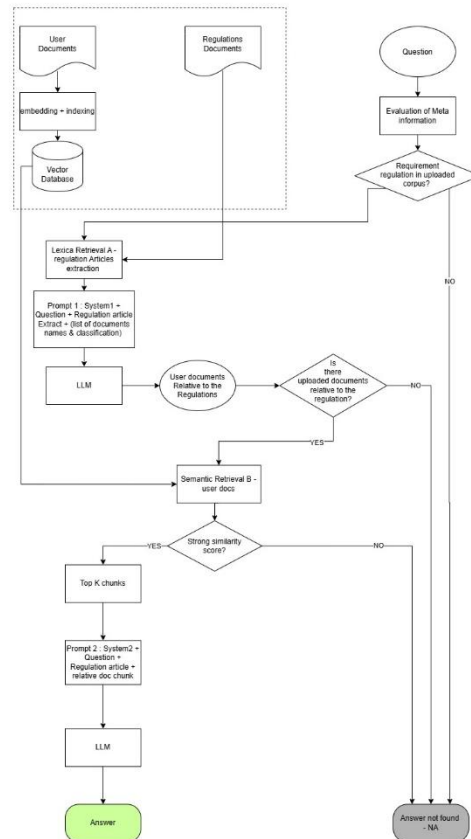
RAG POC – Pipeline

1. Overview

The pipeline answers the compliance questionnaire in `data/questions.json` using the regulation article each question cites as context.

a question never reaches the model without a regulation article and at least one relevant document chunk.

A full run costs at most **2 chat calls** (OpenRouter) and **1 embedding call** (Gemini) per question; `answer_all()` repeats it for every question.



Where each box lives

- `rag/pipeline.py` : `answer_question()` orchestrates the flow; **Answer / NA**
- `rag/indexing.py` : **embedding + indexing** → **Vector Database**
- `rag/questions.py` : **Evaluation of Meta information**
- `rag/regulations.py` : **regulation in corpus?** and **Lexical Retrieval A**
- `rag/prompts.py` : **Prompt 1** (SYSTEM_ROUTER) and **Prompt 2** (SYSTEM_ANSWER)
- `rag/llm.py` : the two **LLM** boxes (`ask()`, `parse_json_reply()`)
- `rag/retrieval.py` : **Semantic Retrieval B** and **Strong similarity score?**
- `api/documentation.py`, `api/rag.py` : upload endpoint and one endpoint per stage

Key parameters (`rag/config.py`, overridable in `.env`)

- `CHUNK_SIZE 512 · CHUNK_OVERLAP 64 · TOP_K 5 · SIMILARITY_CUTOFF 0.62 · EMBED_MODEL gemini-embedding-001 · USE_DOC_ROUTER true · DOC_PREVIEW_CHARS 300 · chat model = inception/mercury-2 via OpenRouter` (cheaper for tests than gpt 4 mini)

How a question can end

- **NA** : regulation file missing (no call made)
- **NA** : cited article not found in the regulation (no call made)
- **NA** : no documents uploaded (no call) or Prompt 1 selects none (1 call)
- **NA** : no chunk above the similarity cutoff (1 call + 1 embedding)
- **answered** : model JSON with YES / NO / MAYBE / ABSTAIN (2 calls + 1 embedding)